

Hilbert-Style Exercise Set for $N(A, B) := \neg(A \rightarrow \neg B)$

Executive summary

All 400 exercises are supplied in one list ordered by MP depth. Within each MP-depth tier, the original relative order is preserved. The achieved distinct-formula count remains 399 because the two preserved contraposition exercises are identical. The milestone is now exercise 327.

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Exercises

1.

$$A \rightarrow B \rightarrow A$$

2.

$$A \rightarrow A$$

3.

$$(A \rightarrow B) \rightarrow (C \rightarrow A) \rightarrow C \rightarrow B$$

4.

$$(A \rightarrow B \rightarrow C) \rightarrow B \rightarrow A \rightarrow C$$

5.

$$A \rightarrow \neg\neg A$$

6.

$$\neg\neg\neg A \rightarrow \neg A$$

7.

$$(A \rightarrow B) \rightarrow (\neg B \rightarrow \neg A)$$

8.

$$(A \rightarrow \neg B) \rightarrow (B \rightarrow \neg A)$$

9.

$$A \rightarrow N(A, A)$$

10.

$$B \rightarrow N(B, B)$$

11.

$$C \rightarrow N(C, C)$$

12.

$$D \rightarrow N(D, D)$$

13.

$$N(A, A) \rightarrow A$$

14.

$$N(A, B) \rightarrow N(N(A, B), N(A, B))$$

15.

$$N(A, C) \rightarrow A$$

16.

$$N(A, C) \rightarrow C$$

17.

$$N(A, C) \rightarrow N(C, A)$$

18.

$$N(A, D) \rightarrow A$$

19.

$$N(A, D) \rightarrow D$$

20.

$$N(A, D) \rightarrow N(D, A)$$

21.

$$N(A, N(B, C)) \rightarrow A$$

22.

$$N(A, N(B, C)) \rightarrow N(N(B, C), A)$$

23.

$$N(B, B) \rightarrow B$$

24.

$$N(B, C) \rightarrow B$$

25.

$$N(B, C) \rightarrow C$$

26.

$$N(B, C) \rightarrow N(C, B)$$

27.

$$N(B, D) \rightarrow B$$

28.

$$N(B, D) \rightarrow D$$

29.

$$N(B, D) \rightarrow N(D, B)$$

30.

$$N(C, C) \rightarrow C$$

31.

$$N(C, D) \rightarrow C$$

32.

$$N(C, D) \rightarrow D$$

33.

$$N(C, D) \rightarrow N(D, C)$$

34.

$$N(D, D) \rightarrow D$$

35.

$$N(N(A, B), C) \rightarrow C$$

36.

$$N(N(A, B), C) \rightarrow N(C, N(A, B))$$

37.

$$N(N(A, B), N(A, B)) \rightarrow N(A, B)$$

38.

$$N(N(A, B), N(C, D)) \rightarrow N(A, B)$$

39.

$$N(N(A, B), N(C, D)) \rightarrow N(C, D)$$

40.

$$N(N(A, B), N(C, D)) \rightarrow N(N(C, D), N(A, B))$$

41.

$$N(\neg A, B) \rightarrow B$$

42.

$$N(\neg A, B) \rightarrow N(B, \neg A)$$

43.

$$N(\neg A, B) \rightarrow \neg A$$

44.

$$(A \rightarrow B) \rightarrow (\neg\neg A \rightarrow \neg\neg B)$$

45.

$$(\neg A \rightarrow B) \rightarrow (\neg B \rightarrow \neg\neg A)$$

46.

$$(A \rightarrow B \rightarrow C) \rightarrow (\neg C \rightarrow A \rightarrow \neg B)$$

47.

$$(A \rightarrow B \rightarrow \neg C) \rightarrow (C \rightarrow A \rightarrow \neg B)$$

48.

$$A \rightarrow C \rightarrow N(A, C)$$

49.

$$A \rightarrow D \rightarrow N(A, D)$$

50.

$$A \rightarrow N(B, C) \rightarrow N(A, N(B, C))$$

51.

$$B \rightarrow C \rightarrow N(B, C)$$

52.

$$B \rightarrow D \rightarrow N(B, D)$$

53.

$$C \rightarrow D \rightarrow N(C, D)$$

54.

$$N(A, B) \rightarrow C \rightarrow N(N(A, B), C)$$

55.

$$N(A, B) \rightarrow N(C, D) \rightarrow N(N(A, B), N(C, D))$$

56.

$$N(A, C) \rightarrow \neg\neg A$$

57.

$$N(A, C) \rightarrow \neg\neg C$$

58.

$$N(A, C) \rightarrow \neg\neg N(A, C)$$

59.

$$N(A, D) \rightarrow \neg\neg A$$

60.

$$N(A, D) \rightarrow \neg\neg D$$

61.

$$N(A, D) \rightarrow \neg\neg N(A, D)$$

62.

$$N(A, N(B, C)) \rightarrow \neg\neg N(A, N(B, C))$$

63.

$$N(A, N(B, C)) \rightarrow \neg\neg N(B, C)$$

64.

$$N(B, C) \rightarrow \neg\neg B$$

65.

$$N(B, C) \rightarrow \neg\neg C$$

66.

$$N(B, C) \rightarrow \neg\neg N(B, C)$$

67.

$$N(B, D) \rightarrow \neg\neg B$$

68.

$$N(B, D) \rightarrow \neg\neg D$$

69.

$$N(B, D) \rightarrow \neg\neg N(B, D)$$

70.

$$N(C, D) \rightarrow \neg\neg C$$

71.

$$N(C, D) \rightarrow \neg\neg D$$

72.

$$N(C, D) \rightarrow \neg\neg N(C, D)$$

73.

$$N(N(A, B), C) \rightarrow \neg\neg N(A, B)$$

74.

$$N(N(A, B), C) \rightarrow \neg\neg N(N(A, B), C)$$

75.

$$N(N(A, B), N(C, D)) \rightarrow \neg\neg N(A, B)$$

76.

$$N(N(A, B), N(C, D)) \rightarrow \neg\neg N(C, D)$$

77.

$$N(N(A, B), N(C, D)) \rightarrow \neg\neg N(N(A, B), N(C, D))$$

78.

$$N(\neg A, B) \rightarrow \neg\neg B$$

79.

$$N(\neg A, B) \rightarrow \neg\neg N(\neg A, B)$$

80.

$$N(\neg A, B) \rightarrow \neg\neg\neg A$$

81.

$$\neg A \rightarrow B \rightarrow N(\neg A, B)$$

82.

$$A \rightarrow B \rightarrow N(A, B)$$

83.

$$\neg A \rightarrow A \rightarrow \neg B$$

84.

$$\neg B \rightarrow A \rightarrow \neg B$$

85.

$$(A \rightarrow C) \rightarrow \neg C \rightarrow A \rightarrow \neg B$$

86.

$$(B \rightarrow C) \rightarrow \neg C \rightarrow A \rightarrow \neg B$$

87. $(A \rightarrow C) \rightarrow (B \rightarrow D) \rightarrow (C \rightarrow \neg D) \rightarrow A \rightarrow \neg B$
88. $(B \rightarrow \neg A) \rightarrow (A \rightarrow \neg B)$
89. $(A \rightarrow \neg B) \rightarrow (B \rightarrow \neg A)$
90. $(A \rightarrow \neg B) \rightarrow \neg\neg(A \rightarrow \neg B)$
91. $N(A, B) \rightarrow \neg\neg N(A, B)$
92. $\neg\neg N(A, B) \rightarrow N(A, B)$
93. $(A \rightarrow \neg B) \rightarrow \neg N(A, B)$
94. $N(A, B) \rightarrow N(A, N(A, B))$
95. $N(A, C) \rightarrow N(A, N(A, C))$
96. $N(A, N(A, B)) \rightarrow N(A, B)$
97. $N(A, N(A, C)) \rightarrow N(A, C)$
98. $N(A, N(B, C)) \rightarrow N(N(A, B), C)$
99. $N(A, N(B, C)) \rightarrow N(N(A, B), N(A, C))$
100. $N(A, N(B, D)) \rightarrow N(N(A, B), D)$
101. $N(A, N(B, D)) \rightarrow N(N(A, B), N(A, D))$

102. $N(A, N(B, N(C, D))) \rightarrow N(N(A, B), N(A, N(C, D)))$
103. $N(A, N(B, N(C, D))) \rightarrow N(N(A, B), N(C, D))$
104. $N(A, N(C, D)) \rightarrow N(N(A, C), D)$
105. $N(A, N(C, D)) \rightarrow N(N(A, C), N(A, D))$
106. $N(A, N(N(B, C), D)) \rightarrow N(N(A, N(B, C)), D)$
107. $N(A, N(N(B, C), D)) \rightarrow N(N(A, N(B, C)), N(A, D))$
108. $N(B, C) \rightarrow N(B, N(B, C))$
109. $N(B, N(B, C)) \rightarrow N(B, C)$
110. $N(B, N(C, D)) \rightarrow N(N(B, C), D)$
111. $N(B, N(C, D)) \rightarrow N(N(B, C), N(B, D))$
112. $N(N(A, B), C) \rightarrow N(A, N(B, C))$
113. $N(N(A, B), C) \rightarrow N(N(A, B), N(N(A, B), C))$
114. $N(N(A, B), D) \rightarrow N(A, N(B, D))$
115. $N(N(A, B), N(A, C)) \rightarrow N(A, N(B, C))$
116. $N(N(A, B), N(A, D)) \rightarrow N(A, N(B, D))$

117. $N(N(A, B), N(A, N(C, D))) \rightarrow N(A, N(B, N(C, D)))$
118. $N(N(A, B), N(C, D)) \rightarrow N(A, N(B, N(C, D)))$
119. $N(N(A, B), N(C, D)) \rightarrow N(N(N(A, B), C), D)$
120. $N(N(A, B), N(C, D)) \rightarrow N(N(N(A, B), C), N(N(A, B), D))$
121. $N(N(A, B), N(N(A, B), C)) \rightarrow N(N(A, B), C)$
122. $N(N(A, C), D) \rightarrow N(A, N(C, D))$
123. $N(N(A, C), N(A, D)) \rightarrow N(A, N(C, D))$
124. $N(N(A, N(B, C)), D) \rightarrow N(A, N(N(B, C), D))$
125. $N(N(A, N(B, C)), N(A, D)) \rightarrow N(A, N(N(B, C), D))$
126. $N(N(B, C), D) \rightarrow N(B, N(C, D))$
127. $N(N(B, C), N(B, D)) \rightarrow N(B, N(C, D))$
128. $N(N(N(A, B), C), D) \rightarrow N(N(A, B), N(C, D))$
129. $N(N(N(A, B), C), N(N(A, B), D)) \rightarrow N(N(A, B), N(C, D))$
130. $N(N(\neg A, B), C) \rightarrow N(\neg A, N(B, C))$
131. $N(N(\neg A, B), N(\neg A, C)) \rightarrow N(\neg A, N(B, C))$

132.
$$N(\neg A, N(B, C)) \rightarrow N(N(\neg A, B), C)$$
133.
$$N(\neg A, N(B, C)) \rightarrow N(N(\neg A, B), N(\neg A, C))$$
134.
$$N(A, B) \rightarrow \neg\neg B$$
135.
$$N(A, B) \rightarrow \neg\neg A$$
136.
$$N(A, B) \rightarrow A \rightarrow \neg\neg B$$
137.
$$N(A, B) \rightarrow B \rightarrow \neg\neg A$$
138.
$$N(A, B) \rightarrow (\neg A \rightarrow C)$$
139.
$$N(A, B) \rightarrow (\neg B \rightarrow C)$$
140.
$$N(A, A) \rightarrow \neg\neg A$$
141.
$$\neg\neg A \rightarrow N(A, A)$$
142.
$$N(A, B) \rightarrow N(A, A)$$
143.
$$N(A, B) \rightarrow N(B, B)$$
144.
$$N(A, B) \rightarrow N(B, A)$$
145.
$$N(B, A) \rightarrow N(A, B)$$
146.
$$((A \rightarrow B) \rightarrow C) \rightarrow N(A \rightarrow B, D) \rightarrow C$$

147. $((A \rightarrow B) \rightarrow C) \rightarrow N(A \rightarrow B, D) \rightarrow N(C, D)$
148. $((A \rightarrow B) \rightarrow C) \rightarrow N(D, A \rightarrow B) \rightarrow C$
149. $((A \rightarrow B) \rightarrow C) \rightarrow N(D, A \rightarrow B) \rightarrow N(D, C)$
150. $(A \rightarrow B) \rightarrow N(A, C) \rightarrow B$
151. $(A \rightarrow B) \rightarrow N(A, C) \rightarrow N(B, C)$
152. $(A \rightarrow B) \rightarrow N(A, N(C, D)) \rightarrow B$
153. $(A \rightarrow B) \rightarrow N(A, N(C, D)) \rightarrow N(B, N(C, D))$
154. $(A \rightarrow B) \rightarrow N(C, A) \rightarrow B$
155. $(A \rightarrow B) \rightarrow N(C, A) \rightarrow N(C, B)$
156. $(A \rightarrow B) \rightarrow N(N(C, D), A) \rightarrow B$
157. $(A \rightarrow B) \rightarrow N(N(C, D), A) \rightarrow N(N(C, D), B)$
158. $(A \rightarrow C) \rightarrow N(B, A) \rightarrow C$
159. $(A \rightarrow C) \rightarrow N(B, A) \rightarrow N(B, C)$
160. $(A \rightarrow N(B, C)) \rightarrow N(A, D) \rightarrow N(B, C)$
161. $(A \rightarrow N(B, C)) \rightarrow N(A, D) \rightarrow N(N(B, C), D)$

162. $(A \rightarrow N(B, C)) \rightarrow N(D, A) \rightarrow N(B, C)$
163. $(A \rightarrow N(B, C)) \rightarrow N(D, A) \rightarrow N(D, N(B, C))$
164. $(A \rightarrow \neg B) \rightarrow N(A, C) \rightarrow N(\neg B, C)$
165. $(A \rightarrow \neg B) \rightarrow N(A, C) \rightarrow \neg B$
166. $(A \rightarrow \neg B) \rightarrow N(C, A) \rightarrow N(C, \neg B)$
167. $(A \rightarrow \neg B) \rightarrow N(C, A) \rightarrow \neg B$
168. $(B \rightarrow C) \rightarrow N(A, B) \rightarrow N(A, C)$
169. $(B \rightarrow C) \rightarrow N(B, A) \rightarrow C$
170. $(B \rightarrow C) \rightarrow N(B, A) \rightarrow N(C, A)$
171. $(N(A, B) \rightarrow C) \rightarrow N(D, N(A, B)) \rightarrow C$
172. $(N(A, B) \rightarrow C) \rightarrow N(D, N(A, B)) \rightarrow N(D, C)$
173. $(N(A, B) \rightarrow C) \rightarrow N(N(A, B), D) \rightarrow C$
174. $(N(A, B) \rightarrow C) \rightarrow N(N(A, B), D) \rightarrow N(C, D)$
175. $(N(A, B) \rightarrow N(C, D)) \rightarrow N(E, N(A, B)) \rightarrow N(C, D)$
176. $(N(A, B) \rightarrow N(C, D)) \rightarrow N(E, N(A, B)) \rightarrow N(E, N(C, D))$

177. $(N(A, B) \rightarrow N(C, D)) \rightarrow N(N(A, B), E) \rightarrow N(C, D)$
178. $(N(A, B) \rightarrow N(C, D)) \rightarrow N(N(A, B), E) \rightarrow N(N(C, D), E)$
179. $(\neg A \rightarrow B) \rightarrow N(C, \neg A) \rightarrow B$
180. $(\neg A \rightarrow B) \rightarrow N(C, \neg A) \rightarrow N(C, B)$
181. $(\neg A \rightarrow B) \rightarrow N(\neg A, C) \rightarrow B$
182. $(\neg A \rightarrow B) \rightarrow N(\neg A, C) \rightarrow N(B, C)$
183. $(A \rightarrow C) \rightarrow (N(A, B) \rightarrow N(C, B))$
184. $(B \rightarrow D) \rightarrow (N(A, B) \rightarrow N(A, D))$
185. $(A \rightarrow C) \rightarrow (B \rightarrow D) \rightarrow (N(A, B) \rightarrow N(C, D))$
186. $N(A, B) \rightarrow N(\neg\neg A, B)$
187. $N(A, B) \rightarrow N(A, \neg\neg B)$
188. $N(A, B) \rightarrow N(\neg\neg A, \neg\neg B)$
189. $N(A, B) \rightarrow N(\neg\neg B, \neg\neg A)$
190. $N(A, B) \rightarrow N(N(A, A), B)$
191. $N(A, B) \rightarrow N(A, N(B, B))$

192.

$$N(A, B) \rightarrow N(N(A, A), N(B, B))$$

193.

$$N(\neg\neg A, \neg\neg B) \rightarrow N(A, B)$$

194.

$$N(N(A, A), N(B, B)) \rightarrow N(A, B)$$

195.

$$((A \rightarrow B) \rightarrow C) \rightarrow (D \rightarrow E) \rightarrow N(A \rightarrow B, D) \rightarrow N(C, E)$$

196.

$$(A \rightarrow B \rightarrow D) \rightarrow N(A, B) \rightarrow D$$

197.

$$(A \rightarrow B) \rightarrow (B \rightarrow C) \rightarrow N(A, B) \rightarrow N(B, C)$$

198.

$$(A \rightarrow C \rightarrow D) \rightarrow N(A, C) \rightarrow D$$

199.

$$(A \rightarrow C) \rightarrow (\neg B \rightarrow \neg D) \rightarrow N(A, \neg B) \rightarrow N(C, \neg D)$$

200.

$$(A \rightarrow N(A, C)) \rightarrow (B \rightarrow N(B, D)) \rightarrow N(A, B) \rightarrow N(N(A, C), N(B, D))$$

201.

$$(A \rightarrow N(B, C) \rightarrow D) \rightarrow N(A, N(B, C)) \rightarrow D$$

202.

$$(A \rightarrow \neg B \rightarrow C) \rightarrow N(A, \neg B) \rightarrow C$$

203.

$$(A \rightarrow \neg C) \rightarrow N(A, \neg B) \rightarrow \neg C$$

204.

$$(A \rightarrow \neg D) \rightarrow N(A, B) \rightarrow \neg D$$

205.

$$(A \rightarrow \neg D) \rightarrow N(A, C) \rightarrow \neg D$$

206.

$$(A \rightarrow \neg D) \rightarrow N(A, N(B, C)) \rightarrow \neg D$$

207.

$$(B \rightarrow C \rightarrow D) \rightarrow N(B, C) \rightarrow D$$

208.

$$(B \rightarrow C) \rightarrow (A \rightarrow D) \rightarrow N(B, A) \rightarrow N(C, D)$$

209.

$$(B \rightarrow \neg D) \rightarrow N(B, C) \rightarrow \neg D$$

210.

$$(N(A, B) \rightarrow C \rightarrow D) \rightarrow N(N(A, B), C) \rightarrow D$$

211.

$$(N(A, B) \rightarrow C) \rightarrow (N(C, D) \rightarrow E) \rightarrow N(N(A, B), N(C, D)) \rightarrow N(C, E)$$

212.

$$(N(A, B) \rightarrow C) \rightarrow A \rightarrow B \rightarrow C$$

213.

$$(N(A, B) \rightarrow D) \rightarrow A \rightarrow B \rightarrow D$$

214.

$$(N(A, B) \rightarrow \neg D) \rightarrow N(N(A, B), C) \rightarrow \neg D$$

215.

$$(N(A, C) \rightarrow D) \rightarrow A \rightarrow C \rightarrow D$$

216.

$$(N(A, N(B, C)) \rightarrow D) \rightarrow A \rightarrow N(B, C) \rightarrow D$$

217.

$$(N(A, \neg B) \rightarrow C) \rightarrow A \rightarrow \neg B \rightarrow C$$

218.

$$(N(B, C) \rightarrow D) \rightarrow B \rightarrow C \rightarrow D$$

219.

$$(N(N(A, B), C) \rightarrow D) \rightarrow N(A, B) \rightarrow C \rightarrow D$$

220.

$$(N(\neg A, B) \rightarrow C) \rightarrow \neg A \rightarrow B \rightarrow C$$

221.

$$(\neg A \rightarrow B \rightarrow C) \rightarrow N(\neg A, B) \rightarrow C$$

222. $(\neg A \rightarrow \neg B) \rightarrow (B \rightarrow C) \rightarrow N(\neg A, B) \rightarrow N(\neg B, C)$
223. $(\neg A \rightarrow \neg C) \rightarrow N(\neg A, B) \rightarrow \neg C$
224. $(A \rightarrow C) \rightarrow (N(A, B) \rightarrow \neg\neg C)$
225. $(B \rightarrow C) \rightarrow (N(A, B) \rightarrow \neg\neg C)$
226. $(A \rightarrow B \rightarrow C) \rightarrow (N(A, B) \rightarrow \neg\neg C)$
227. $N(A, B) \rightarrow (A \rightarrow C) \rightarrow \neg\neg C$
228. $N(A, B) \rightarrow (B \rightarrow C) \rightarrow \neg\neg C$
229. $N(A, B) \rightarrow (A \rightarrow B \rightarrow C) \rightarrow \neg\neg C$
230. $(A \rightarrow \neg C) \rightarrow (N(A, B) \rightarrow \neg C)$
231. $(B \rightarrow \neg C) \rightarrow (N(A, B) \rightarrow \neg C)$
232. $(A \rightarrow B \rightarrow \neg C) \rightarrow (N(A, B) \rightarrow \neg C)$
233. $N(A, B) \rightarrow (A \rightarrow B \rightarrow \neg C) \rightarrow \neg C$
234. $(A \rightarrow B \rightarrow C) \rightarrow (\neg\neg C \rightarrow C) \rightarrow (N(A, B) \rightarrow C)$
235. $(\neg\neg C \rightarrow C) \rightarrow (A \rightarrow B \rightarrow C) \rightarrow (N(A, B) \rightarrow C)$
236. $(A \rightarrow \neg C) \rightarrow \neg N(A, C)$

237.

$$(A \rightarrow \neg D) \rightarrow \neg N(A, D)$$

238.

$$(A \rightarrow \neg N(B, C)) \rightarrow \neg N(A, N(B, C))$$

239.

$$(A \rightarrow \neg\neg B) \rightarrow \neg N(A, \neg B)$$

240.

$$(B \rightarrow \neg C) \rightarrow \neg N(B, C)$$

241.

$$(B \rightarrow \neg D) \rightarrow \neg N(B, D)$$

242.

$$(C \rightarrow \neg D) \rightarrow \neg N(C, D)$$

243.

$$(N(A, B) \rightarrow \neg C) \rightarrow \neg N(N(A, B), C)$$

244.

$$(\neg A \rightarrow \neg B) \rightarrow \neg N(\neg A, B)$$

245.

$$N(A, \neg A) \rightarrow C$$

246.

$$N(A, \neg A) \rightarrow D$$

247.

$$N(A, \neg A) \rightarrow N(B, C)$$

248.

$$N(A, \neg A) \rightarrow N(B, D)$$

249.

$$N(A, \neg A) \rightarrow N(N(B, C), D)$$

250.

$$N(B, \neg B) \rightarrow A$$

251.

$$N(B, \neg B) \rightarrow N(C, A)$$

252.
$$N(N(A, B), \neg N(A, B)) \rightarrow D$$
253.
$$N(N(A, B), \neg N(A, B)) \rightarrow N(C, D)$$
254.
$$\neg N(A, B) \rightarrow A \rightarrow \neg B$$
255.
$$\neg N(A, C) \rightarrow A \rightarrow \neg C$$
256.
$$\neg N(A, D) \rightarrow A \rightarrow \neg D$$
257.
$$\neg N(A, N(B, C)) \rightarrow A \rightarrow \neg N(B, C)$$
258.
$$\neg N(A, \neg B) \rightarrow A \rightarrow \neg \neg B$$
259.
$$\neg N(B, C) \rightarrow B \rightarrow \neg C$$
260.
$$\neg N(B, D) \rightarrow B \rightarrow \neg D$$
261.
$$\neg N(C, D) \rightarrow C \rightarrow \neg D$$
262.
$$\neg N(N(A, B), C) \rightarrow N(A, B) \rightarrow \neg C$$
263.
$$\neg N(\neg A, B) \rightarrow \neg A \rightarrow \neg B$$
264.
$$\neg \neg N(A, C) \rightarrow N(A, C)$$
265.
$$\neg \neg N(A, D) \rightarrow N(A, D)$$
266.
$$\neg \neg N(A, N(B, C)) \rightarrow N(A, N(B, C))$$

267. $\neg\neg N(A, \neg B) \rightarrow N(A, \neg B)$
268. $\neg\neg N(B, C) \rightarrow N(B, C)$
269. $\neg\neg N(B, D) \rightarrow N(B, D)$
270. $\neg\neg N(C, D) \rightarrow N(C, D)$
271. $\neg\neg N(N(A, B), C) \rightarrow N(N(A, B), C)$
272. $\neg\neg N(\neg A, B) \rightarrow N(\neg A, B)$
273. $(A \& B) \rightarrow N(A, B)$
274. $N(A, B) \rightarrow \neg\neg(A \& B)$
275. $\neg\neg(A \& B) \rightarrow N(A, B)$
276. $N(A, B) \rightarrow (\neg\neg A \& \neg\neg B)$
277. $(\neg\neg A \& \neg\neg B) \rightarrow N(A, B)$
278. $N(N(A, B), C) \rightarrow N(A, B)$
279. $N(A, N(B, C)) \rightarrow N(B, C)$
280. $N(N(A, B), C) \rightarrow \neg\neg C$
281. $N(A, N(B, C)) \rightarrow \neg\neg A$

282.
$$N(N(A, B), N(C, D)) \rightarrow N(A, C)$$
283.
$$N(N(A, B), N(C, D)) \rightarrow N(B, D)$$
284.
$$N(N(A, B), N(C, D)) \rightarrow N(N(A, C), N(B, D))$$
285.
$$N(N(A, B), N(C, D)) \rightarrow N(A, D)$$
286.
$$N(N(A, B), N(C, D)) \rightarrow N(B, C)$$
287.
$$N(N(A, B), N(\neg C, D)) \rightarrow N(A, D)$$
288.
$$N(N(A, B), N(\neg C, D)) \rightarrow N(A, \neg C)$$
289.
$$N(N(A, B), N(\neg C, D)) \rightarrow N(B, D)$$
290.
$$N(N(A, B), N(\neg C, D)) \rightarrow N(B, \neg C)$$
291.
$$N(N(A, B), N(\neg C, D)) \rightarrow N(N(A, \neg C), N(B, D))$$
292.
$$N(N(A, C), N(B, D)) \rightarrow N(A, B)$$
293.
$$N(N(A, C), N(B, D)) \rightarrow N(A, D)$$
294.
$$N(N(A, C), N(B, D)) \rightarrow N(C, B)$$
295.
$$N(N(A, C), N(B, D)) \rightarrow N(C, D)$$
296.
$$N(N(A, C), N(B, D)) \rightarrow N(N(A, B), N(C, D))$$

297.
$$N(N(A, D), N(B, C)) \rightarrow N(A, B)$$
298.
$$N(N(A, D), N(B, C)) \rightarrow N(A, C)$$
299.
$$N(N(A, D), N(B, C)) \rightarrow N(D, B)$$
300.
$$N(N(A, D), N(B, C)) \rightarrow N(D, C)$$
301.
$$N(N(A, D), N(B, C)) \rightarrow N(N(A, B), N(D, C))$$
302.
$$N(N(A, N(B, C)), N(D, E)) \rightarrow N(A, D)$$
303.
$$N(N(A, N(B, C)), N(D, E)) \rightarrow N(A, E)$$
304.
$$N(N(A, N(B, C)), N(D, E)) \rightarrow N(N(A, D), N(N(B, C), E))$$
305.
$$N(N(A, N(B, C)), N(D, E)) \rightarrow N(N(B, C), D)$$
306.
$$N(N(A, N(B, C)), N(D, E)) \rightarrow N(N(B, C), E)$$
307.
$$N(N(A, \neg B), N(C, D)) \rightarrow N(A, C)$$
308.
$$N(N(A, \neg B), N(C, D)) \rightarrow N(A, D)$$
309.
$$N(N(A, \neg B), N(C, D)) \rightarrow N(N(A, C), N(\neg B, D))$$
310.
$$N(N(A, \neg B), N(C, D)) \rightarrow N(\neg B, C)$$
311.
$$N(N(A, \neg B), N(C, D)) \rightarrow N(\neg B, D)$$

312.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(C, D)$$
313.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(C, E)$$
314.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(A, B), D)$$
315.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(A, B), E)$$
316.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(N(A, B), D), N(C, E))$$
317.
$$N(N(\neg A, B), N(C, D)) \rightarrow N(B, C)$$
318.
$$N(N(\neg A, B), N(C, D)) \rightarrow N(B, D)$$
319.
$$N(N(\neg A, B), N(C, D)) \rightarrow N(N(\neg A, C), N(B, D))$$
320.
$$N(N(\neg A, B), N(C, D)) \rightarrow N(\neg A, C)$$
321.
$$N(N(\neg A, B), N(C, D)) \rightarrow N(\neg A, D)$$
322.
$$\neg\neg A \rightarrow A$$
323.
$$N(A, B) \rightarrow A$$
324.
$$N(A, B) \rightarrow B$$
325.
$$(A \rightarrow C) \rightarrow (N(A, B) \rightarrow C)$$
326.
$$(B \rightarrow C) \rightarrow (N(A, B) \rightarrow C)$$

327.

$$(A \rightarrow B \rightarrow C) \rightarrow (N(A, B) \rightarrow C)$$

[MILESTONE]

328.

$$N(A, B) \rightarrow (A \rightarrow B)$$

329.

$$N(N(A, B), N(C, D)) \rightarrow D$$

330.

$$N(A, N(B, N(D, C))) \rightarrow N(N(A, B), N(D, C))$$

331.

$$N(A, N(C, N(B, D))) \rightarrow N(N(A, C), N(B, D))$$

332.

$$N(B, N(A, N(C, D))) \rightarrow N(N(B, A), N(C, D))$$

333.

$$N(C, N(D, N(B, A))) \rightarrow N(N(A, B), N(D, C))$$

334.

$$N(D, N(B, N(C, A))) \rightarrow N(N(A, C), N(B, D))$$

335.

$$N(D, N(C, N(A, B))) \rightarrow N(N(B, A), N(C, D))$$

336.

$$N(D, N(C, N(B, A))) \rightarrow N(N(A, B), N(C, D))$$

337.

$$N(E, N(D, N(C, N(A, B)))) \rightarrow N(N(N(A, B), C), N(D, E))$$

338.

$$N(N(A, B), N(C, D)) \rightarrow N(D, N(C, N(B, A)))$$

339.

$$N(N(A, B), N(C, D)) \rightarrow N(N(A, C), N(D, B))$$

340.

$$N(N(A, B), N(C, D)) \rightarrow N(N(A, D), N(B, C))$$

341.

$$N(N(A, B), N(C, D)) \rightarrow N(N(B, C), N(A, D))$$

342. $N(N(A, B), N(C, D)) \rightarrow N(N(B, D), N(C, A))$
343. $N(N(A, B), N(C, N(D, E))) \rightarrow N(N(N(A, B), C), N(D, E))$
344. $N(N(A, B), N(D, C)) \rightarrow N(A, N(B, N(D, C)))$
345. $N(N(A, B), N(D, C)) \rightarrow N(C, N(D, N(B, A)))$
346. $N(N(A, B), N(D, C)) \rightarrow N(N(A, C), N(B, D))$
347. $N(N(A, B), N(D, C)) \rightarrow N(N(A, D), N(C, B))$
348. $N(N(A, B), N(D, C)) \rightarrow N(N(B, C), N(D, A))$
349. $N(N(A, B), N(D, C)) \rightarrow N(N(B, D), N(A, C))$
350. $N(N(A, C), N(B, D)) \rightarrow N(A, N(C, N(B, D)))$
351. $N(N(A, C), N(B, D)) \rightarrow N(D, N(B, N(C, A)))$
352. $N(N(A, C), N(B, D)) \rightarrow N(N(A, B), N(D, C))$
353. $N(N(A, C), N(B, D)) \rightarrow N(N(A, D), N(C, B))$
354. $N(N(A, C), N(B, D)) \rightarrow N(N(C, B), N(A, D))$
355. $N(N(A, C), N(B, D)) \rightarrow N(N(C, D), N(B, A))$
356. $N(N(B, A), N(C, D)) \rightarrow N(B, N(A, N(C, D)))$

357.
$$N(N(B, A), N(C, D)) \rightarrow N(D, N(C, N(A, B)))$$
358.
$$N(N(B, A), N(C, D)) \rightarrow N(N(A, C), N(B, D))$$
359.
$$N(N(B, A), N(C, D)) \rightarrow N(N(A, D), N(C, B))$$
360.
$$N(N(B, A), N(C, D)) \rightarrow N(N(B, C), N(D, A))$$
361.
$$N(N(B, A), N(C, D)) \rightarrow N(N(B, D), N(A, C))$$
362.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(E, N(D, N(C, N(A, B))))$$
363.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(A, B), N(C, N(D, E)))$$
364.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(C, D), N(N(A, B), E))$$
365.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(C, E), N(D, N(A, B)))$$
366.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(N(A, B), D), N(E, C))$$
367.
$$N(N(N(A, B), C), N(D, E)) \rightarrow N(N(N(A, B), E), N(C, D))$$
368.
$$A \rightarrow (B \rightarrow C) \rightarrow N(A, B \rightarrow C)$$
369.
$$N(A \rightarrow B, A) \rightarrow A$$
370.
$$N(A \rightarrow B, A) \rightarrow A \rightarrow B$$
371.
$$N(A \rightarrow B, A) \rightarrow N(A, A \rightarrow B)$$

372.

$$N(A \rightarrow B, N(A, C)) \rightarrow A \rightarrow B$$

373.

$$N(A \rightarrow B, N(A, C)) \rightarrow N(A, C)$$

374.

$$N(A \rightarrow B, N(A, C)) \rightarrow N(N(A, C), A \rightarrow B)$$

375.

$$N(A \rightarrow N(B, C), A) \rightarrow A$$

376.

$$N(A \rightarrow N(B, C), A) \rightarrow A \rightarrow N(B, C)$$

377.

$$N(A \rightarrow N(B, C), A) \rightarrow N(A, A \rightarrow N(B, C))$$

378.

$$N(A, C) \rightarrow N(B, C) \rightarrow N(N(A, C), N(B, C))$$

379.

$$N(N(A, B \rightarrow C), A) \rightarrow A$$

380.

$$N(N(A, B \rightarrow C), A) \rightarrow N(A, B \rightarrow C)$$

381.

$$N(N(A, B), N(C, N(D, E))) \rightarrow A$$

382.

$$N(N(A, B), N(C, N(D, E))) \rightarrow B$$

383.

$$N(N(A, B), N(C, N(D, E))) \rightarrow C$$

384.

$$N(N(A, B), N(C, N(D, E))) \rightarrow D$$

385.

$$N(N(A, B), N(C, N(D, E))) \rightarrow E$$

386.

$$N(N(A, B), N(C, N(D, E))) \rightarrow N(A, C)$$

387.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(A, E)$$
388.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(A, N(B, N(C, N(D, E))))$$
389.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(A, N(N(B, C), N(D, E)))$$
390.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(B, D)$$
391.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(C, E)$$
392.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(E, N(D, N(C, N(B, A))))$$
393.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(A, B), C)$$
394.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(A, C), N(B, N(D, E)))$$
395.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(A, D), N(B, N(C, E)))$$
396.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(A, E), N(B, N(C, D)))$$
397.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(A, N(B, C)), N(D, E))$$
398.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(B, D), N(A, N(C, E)))$$
399.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(C, E), N(A, N(B, D)))$$
400.
$$N(N(A, B), N(C, N(D, E))) \rightarrow N(N(N(A, B), N(C, D)), E)$$

Exercise-to-MP-depth table

Ex.	MP	Ex.	MP	Ex.	MP	Ex.	MP	Ex.	MP	Ex.	MP	Ex.	MP	Ex.	MP
1	0	51	6	101	7	151	8	201	9	251	10	301	11	351	12
2	2	52	6	102	7	152	8	202	9	252	10	302	11	352	12
3	3	53	6	103	7	153	8	203	9	253	10	303	11	353	12
4	4	54	6	104	7	154	8	204	9	254	10	304	11	354	12
5	5	55	6	105	7	155	8	205	9	255	10	305	11	355	12
6	5	56	6	106	7	156	8	206	9	256	10	306	11	356	12
7	5	57	6	107	7	157	8	207	9	257	10	307	11	357	12
8	5	58	6	108	7	158	8	208	9	258	10	308	11	358	12
9	5	59	6	109	7	159	8	209	9	259	10	309	11	359	12
10	5	60	6	110	7	160	8	210	9	260	10	310	11	360	12
11	5	61	6	111	7	161	8	211	9	261	10	311	11	361	12
12	5	62	6	112	7	162	8	212	9	262	10	312	11	362	12
13	5	63	6	113	7	163	8	213	9	263	10	313	11	363	12
14	5	64	6	114	7	164	8	214	9	264	10	314	11	364	12
15	5	65	6	115	7	165	8	215	9	265	10	315	11	365	12
16	5	66	6	116	7	166	8	216	9	266	10	316	11	366	12
17	5	67	6	117	7	167	8	217	9	267	10	317	11	367	12
18	5	68	6	118	7	168	8	218	9	268	10	318	11	368	13
19	5	69	6	119	7	169	8	219	9	269	10	319	11	369	13
20	5	70	6	120	7	170	8	220	9	270	10	320	11	370	13
21	5	71	6	121	7	171	8	221	9	271	10	321	11	371	13
22	5	72	6	122	7	172	8	222	9	272	10	322	12	372	13
23	5	73	6	123	7	173	8	223	9	273	11	323	12	373	13
24	5	74	6	124	7	174	8	224	10	274	11	324	12	374	13
25	5	75	6	125	7	175	8	225	10	275	11	325	12	375	13
26	5	76	6	126	7	176	8	226	10	276	11	326	12	376	13
27	5	77	6	127	7	177	8	227	10	277	11	327	12	377	13
28	5	78	6	128	7	178	8	228	10	278	11	328	12	378	13
29	5	79	6	129	7	179	8	229	10	279	11	329	12	379	13
30	5	80	6	130	7	180	8	230	10	280	11	330	12	380	13
31	5	81	6	131	7	181	8	231	10	281	11	331	12	381	13
32	5	82	7	132	7	182	8	232	10	282	11	332	12	382	13
33	5	83	7	133	7	183	9	233	10	283	11	333	12	383	13
34	5	84	7	134	8	184	9	234	10	284	11	334	12	384	13
35	5	85	7	135	8	185	9	235	10	285	11	335	12	385	13
36	5	86	7	136	8	186	9	236	10	286	11	336	12	386	13
37	5	87	7	137	8	187	9	237	10	287	11	337	12	387	13
38	5	88	7	138	8	188	9	238	10	288	11	338	12	388	13
39	5	89	7	139	8	189	9	239	10	289	11	339	12	389	13
40	5	90	7	140	8	190	9	240	10	290	11	340	12	390	13
41	5	91	7	141	8	191	9	241	10	291	11	341	12	391	13
42	5	92	7	142	8	192	9	242	10	292	11	342	12	392	13
43	5	93	7	143	8	193	9	243	10	293	11	343	12	393	13
44	6	94	7	144	8	194	9	244	10	294	11	344	12	394	13
45	6	95	7	145	8	195	9	245	10	295	11	345	12	395	13
46	6	96	7	146	8	196	9	246	10	296	11	346	12	396	13
47	6	97	7	147	8	197	9	247	10	297	11	347	12	397	13
48	6	98	7	148	8	198	9	248	10	298	11	348	12	398	13
49	6	99	7	149	8	199	9	249	10	299	11	349	12	399	13
50	6	100	7	150	8	200	9	250	10	300	11	350	12	400	13

Mermaid timeline

```
timeline
    title N(A,B) exercise-set progression
    1 : Exercise set begins
    327 : MILESTONE
    400 : Exercise set completed
```